

INFO 5602
Information Visualization

“Clutter is Your Enemy”

Agenda:

- Clutter + Cognitive load
- Gestalt principles of visual perception (identifying signal and noise)
- Decluttering case study

We are NOT naturally good at **storytelling** and **presentation structuring**

Presentation = Noise + Signal

- *Clutter/distractions on slides*
- *Stakeholder - irrelevant information included*
- *Stakeholder -irrelevant "design" options*
- *Distracting colors, objects included in presentations*
- ***Adds cognitive load to stakeholders***

Find + Remove

- *Stakeholder-relevant information*
- *Stakeholder-relevant charts*
- *Stakeholder-relevant design options*
- *User-centric design for charts*
- ***Improves effective stakeholder communication***

Find/Create + Boost

Anything you add to this *blank* slide adds
cognitive load to an audience

Brief Introduction to the concept of *cognitive load*

- Everything you add to a blank screen takes up cognitive load on the part of your audience – i.e. **takes them brain power to process**
- We experience cognitive load anytime we take in information (especially the first time)
- **Cognitive load** is the mental effort that is required to **learn new information**
 - When we ask a computer to do work, we rely on the computer's processing power (CPU, GPU etc.)
 - When we ask our stakeholders/audience to do work we are leveraging their **mental processing power**
 - Any item in your visualization that **adds no value, increases cognitive load**
- For visualizations, identify anything that is not adding informative value – or is not adding enough informative value to make up for its presence – and remove those things to **reduce cognitive load**

Clutter + Cognitive load

- Clutter are visual elements that take up space **but do not increase understanding on the part of your stakeholders**
- **We should always aim to reduce clutter because it makes our visuals appear more complicated than necessary**
- The presence of clutter in our visual communications:
 - Leads to uncomfortable user (mental) experience for your audience
 - Makes visuals feel more complicated than they actually are
 - Risks our audience **deciding they do not want to take the time to understand what we are showing them** – at which point we have lost our ability to communicate with them

Gestalt principles of visual perception (GPVP)

- GPVP are used to identify which elements in our visuals are **signal** (information we want to communicate) and which might be **noise** (clutter)
- The Gestalt School of Psychology in the 1900s set out to understand how individuals **perceive order in the world around them**
- Gestalt psychology, gestaltism, or configurationism is a school of psychological thought and a theory of perception that emphasizes the processing of **entire patterns and configurations, and not merely individual components.**
 - It emerged in the early twentieth century in Austria and Germany as a rejection of basic principles of Wilhelm Wundt's and Edward Titchener's elementalist and structuralist psychology.

Gestalt principles of visual perception (GPVP)

- GPVP are used to identify which elements in our visuals are **signal** (information we want to communicate) and which might be **noise** (clutter)
- GPVP define how people interact with and create order out of visual stimuli
- We will consider just six principles/laws of GPVP:
 - Proximity
 - Similarity
 - Enclosure
 - Closure
 - Continuity
 - Connection

End of presentation